

3.1 Workflow Overview

All machine learning models follow a general workflow, involving data transformations, model tuning, and model evaluation. The workflow we follow is shown in figure 3.1.

Copied from Hildebrand et al. 2022 (create my own later)

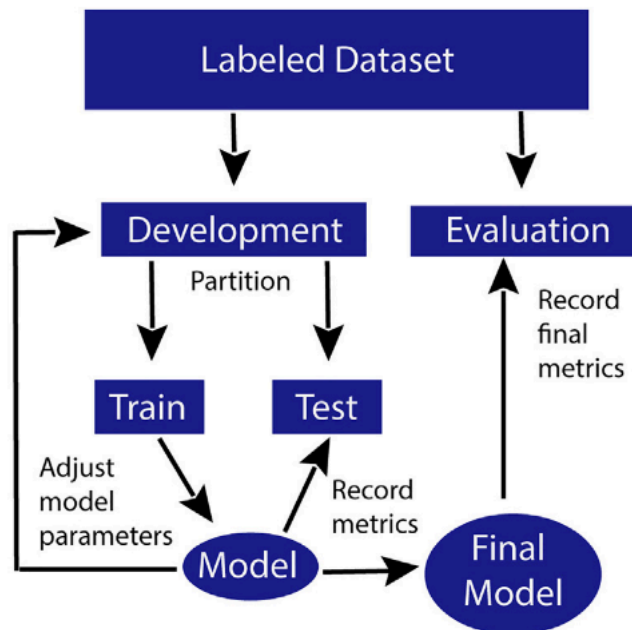


FIGURE 3.1 Workflow

The first step in creating any machine learning model is obtaining data. This data must contain information about the variable of interest. Annotated data provided by Cal Poly's marine science department is our labelled dataset. The labeled dataset is split into two parts, a development data, and evaluation data. The development data will be used in model training and tuning purposes. Evaluation data will be held off until the model is finalized.

The next step is the preprocessing steps. Preprocessing is necessary for audio data, as continuous data is not suited for traditional machine learning models. We preprocess by using Mel Frequency Cepstrum coefficients and context windows.

Model splitting involves using the development data to use in our model tuning. We will use 10-fold-cross validation in our splitting process.

The split development data is used to train our data. We use linear discriminant analysis, k-nearest-neighbors, and random forests as our models of interest. The cross-validated metrics of interest will provide a good sense of how well the model would perform on the evaluation dataset. Model parameters will be tuned with the goal of better metrics. The final model will be the model that produces the best cross-validated metrics. We will focus on recall, precision, F1-score, and the precision-recall curve metrics. The overall success of the model will be determined by the metrics of the final model applied to the evaluation dataset.

3.2 Data Collection

Approximately three hours of whale data was collected off the shore of the central coast of California near ... during the month of ... The hydrophone was submerged at a depth of ... The hydrophone was able to detect frequencies up to 48000 Hz.

Central coast humpback whale migration ...

Throughout this time, 2350 humpback calls were detected and annotated by This was done by simultaneously listening and observing the sound data on the spectrogram.

Figure 3.2 shows a histogram of whale call durations. The average call length lasted 1.634 seconds. The median call length lasted 1.485 seconds. The minimum call length lasted 0.430 seconds, and the maximum call length lasted 10.460 seconds.

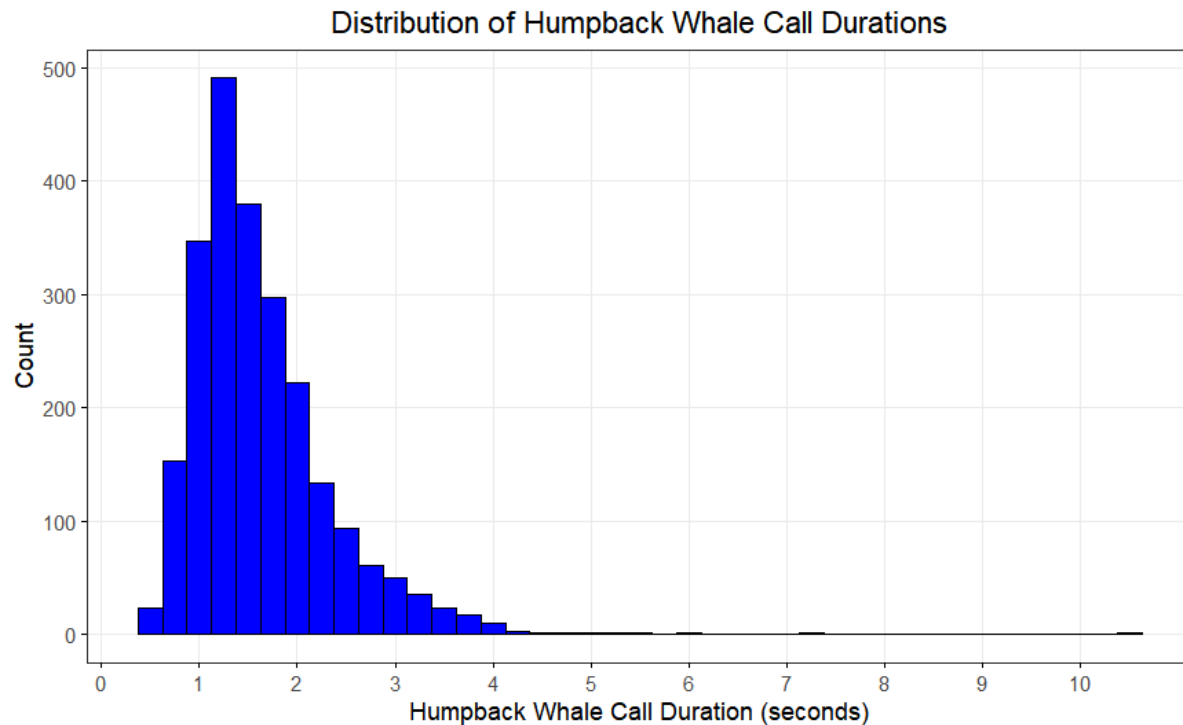


Figure 3.2 Distribution of Humpback Whale Call Durations

Not all whale song vocalizations occurred at independent times. Out of the 2350 observed calls, 158 overlapped with another call.

Based on the data and knowledge from the marine science department (Schroth?), frequencies above 2000 Hz were removed from the data. Excluding higher frequencies allows for the preprocessing steps to focus more on the frequency ranges of interest.

3.3 Preprocessing

Traditional machine learning models are not able to handle sound data without transformation into a tabular form. A large amount of data preprocessing was performed to convert the hydrophone audio recordings to tabular data for machine learning,

Based on the nature of the data collection process, observations are not completely independent. Whale calls will not start perfectly at the beginning of a second and end nicely at

the end of a second. Overlaps of multiple whale vocalizations may also reduce the extracted sound quality of each vocalization. Figure 3.X shows a spectrogram snippet with the corresponding manually annotated humpback whale calls, using the software Raven.

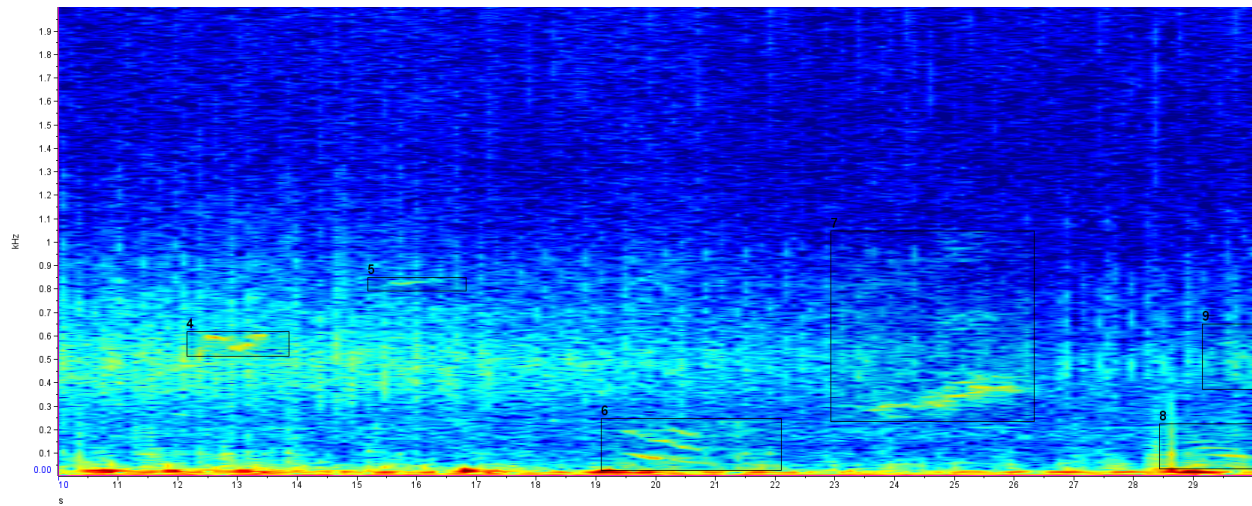


Figure 3.3 A portion of of the spectrogram visualized in Raven

3.3.1 Mel Frequency Cepstrum Coefficients

A necessary aspect of any machine learning model are the predictor variables. Some sort of transformation is necessary to train a machine on the sonically important aspects of a humpback whale vocalization.

Mel-frequency Cepstrum Coefficients are a common method of converting continuous audio data into trainable predictor variables. MFCCs convert audio signals into a scale meant for human perception. Machines and humans do not interpret sound on the same scale. Machines interpret differences in sound linearly. Humans are able to discern well between frequencies in our perceptive range, but do so in a non-linear fashion (Abdul & Al-Talabani 2022). On the Mel scale, equal distances in pitch sound equally distant to a human listener (Dave 2013). Converting a raw audio signal to MFCCs is a multi-step process. The process is shown in Figure 3.4.

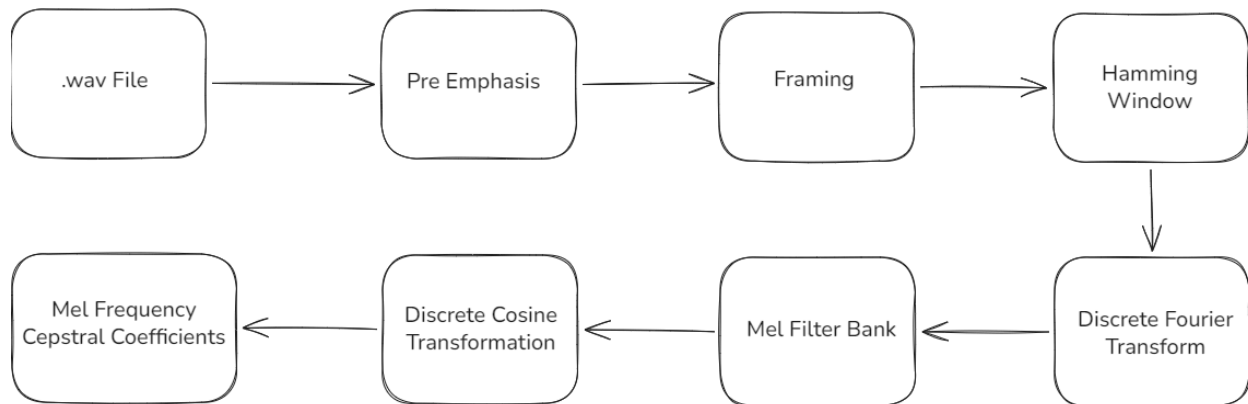


Figure 3.4: The Process of Converting Raw Audio Signal into MFCCs

The first step is pre-emphasis. It emphasizes the quantity in high frequencies to improve the overall signal-to-noise ratio. The pre-emphasis equation is shown in equation 1. The alpha value is the level of emphasis. Standard values are 0.95 or 0.97. We will be using a value of 0.97.

$$y(t) = x(t) - \alpha * x(t - 1) \quad (1)$$

Add less/more about pre-emphasis?

Framing is the process of splitting up the time dimension into windows. A hamming window is applied to each frame. This smooths the edges of time windows to diminish the effect of edges when taking the Discrete Fourier Transform (DFT). The DFT, converts the time domain to the frequency domain, and extracts key frequencies from sound data.

The Mel filter bank helps convert the frequency information into a scale more fit for human perception. An example of the triangular filter bank is shown in Figure 3.4. Typically, 40 Mel-filter banks are used. The Discrete Cosine Transformation (DCT) is the final step to achieve the MFCCs. It “expresses a finite sequence of data points regarding a summation of cosine functions oscillating at different frequencies” (Abdul and Al-Talabani 2022).

Mel-frequency Cepstrum Coefficients are calculated on each time window. For machine learning purposes it is typical to normalize MFCCs. We do so in all of our analyses.

Earlier MFCCs capture the overall spectral shape, while later coefficients capture rapid changes in sound. The figures below are scatterplots of first (Figure 3.2) and twelfth (Figure 3.3) MFCC's over the first 100 seconds.

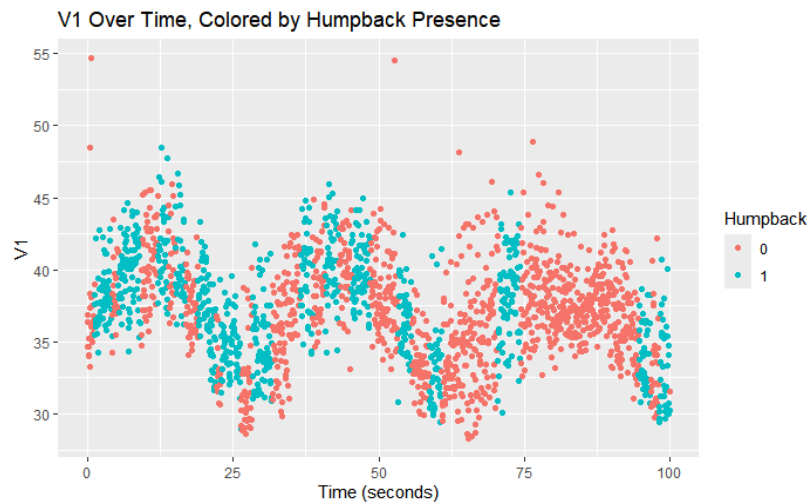


Figure 3.2: The first MFCC over the first 100 seconds of sound

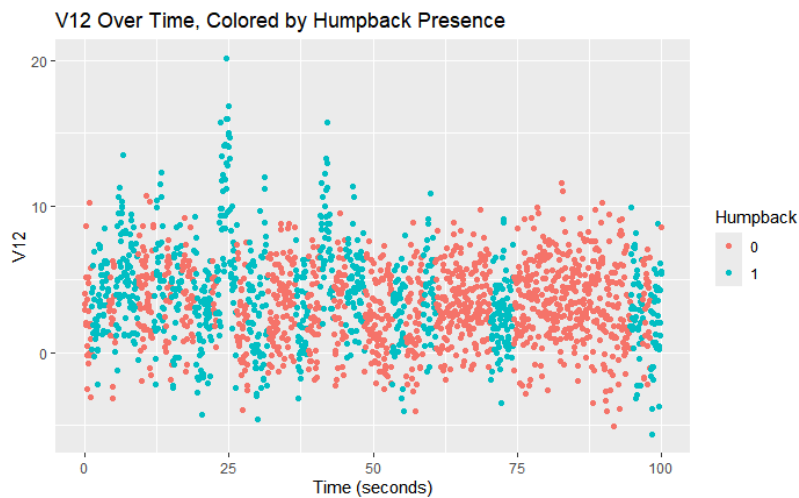


Figure 3.3: The twelfth MFCC over the first 100 seconds of sound

The first MFCC follows a smooth structure with an apparent and predictable wave structure. The twelfth MFCC is more sporadic, with a frequency much faster than that of the first MFCC. Large changes in value occur much more often, possibly due to new sounds being

detected. Later coefficients will be helpful to identify quick changes in sound, such as whale calls. The whale calls occurring at the ~25 second mark and the ~40 second mark have standout values in the 12th MFCC. However, this 12th MFCC may produce false positives due to detected other sounds such as other underwater creatures or boats on the surface. The small group of observations at the ~85 second mark is a noise influx picked up by the hydrophone not caused by a whale.

3.3.2 Gray Whale Project

We further discuss the method used in Wong et al. (2025). In their preprocessing, data was split into 0.1 second long frames. MFCCs were calculated on each frame. Thirteen MFCCs were calculated and used for model training purposes. Each segment was treated as an independent observation used to train and test machine learning models. Their training data structure is shown below in Table 3.1. The thirteen Mel Coefficients are stored in the respective variables (V1 - V13). The ‘Gray Whale’ column is the response column, with 0’s representing no whale presence, and 1’s representing whale presence.

ID	V1	V2	...	V13	Time Start	Time End	Gray Whale
1			...		0	0.1	0/1
2			...		0.1	0.2	0/1
...	...	...	...	...	...	...	...

Table 3.1: Structure of training data for the Capstone Gray Whale Project:

Random forest, k-nearest-neighbors, and an ensemble model of the two were used. Twenty of the thirty-minute-long files were used for model tuning, with another ten being used for evaluation. These machine learning models produced their predictions for each 0.1 second interval.

These individual interval predictions were not the final predictions. An additional aggregation metric was calculated, in the form of a rolling average. A rolling average of length n seconds considered all predictions in the last n seconds. Shorter rolling averages lead to more narrow and frequent prediction intervals. Longer rolling averages result in wider, less frequent prediction intervals.

For their model assessment, they took two rolling average lengths, being ‘wide’ and ‘precise’. The length of the precise intervals are ~ 8 seconds and wide intervals are ~ 30 seconds. The ultimate threshold is not explicitly stated in the paper. Both types of intervals using a random forest model based on MFCC’s resulted in a recall of 0.95. However, the wider intervals had a better precision score of 0.97, compared to the 0.90 from the narrower intervals.

The end of the paper briefly mentions the challenge of humpback whales. Humpback whale calls have a larger variance in call frequencies, and other species are present during humpback whale migration. Additional or different preprocessing methods should be considered to obtain better results. We will apply their method to the provided humpback whale data as well for verification and comparison purposes.

3.3.3 Humpback Whale Additional Preprocessing

The structure used for training, testing, and evaluation did not capture enough about the nature of a humpback whale call. The machine learning models treated each row as independent, which is not ideal when a 0.1 second time segment does not capture humpback whale calls.

We build upon their method to better capture the complexity and diversity of frequencies. We start with similar preprocessing steps as the capstone project. First, we split the sound data into 0.1 second windows. However we use a 0.05 second overlap instead of no overlap. A benefit

of the overlap is to minimize the amount of data lost due to unfortunate cutoff spacing. Next, we apply MFCCs to segmented sound data.

3.3.4 Context Windows

Context windows are an extra step of preprocessing beyond that of Wong et al. (2025). A context window to a fixed length of data used for training (Allen et al. 2021). The goal is to effectively capture the time nature of a call. Windows too small will lead to the issue described above, in which the machine learning model could not identify a humpback whale based on only a small portion of a call. Windows too large would lead to higher odds of multiple whale calls being present in a window, which is not ideal for training purposes. Large windows will also lead to more noise surrounding a whale call, reducing the ability for the machine learning model to hone in on the whale call in the window.

For training and testing purposes context windows are considered to be positive if any whale call is present. If no humpback whale call is present within the window, then the window is treated as a negative. A good context window length provides a good intersection between effectively capturing a typical Humpback whale call, while minimizing the frequency of capturing more than one Humpback whale call. Context windows in studies have differed in length. Factors to consider for the length of are the typical length of a whale call, how often whale calls overlap, and the overall sparsity of calls.

Context windows have been modified slightly from other papers. Other papers avoided the overlapping of context windows, and instead used a time-shift augmentation (Kather et al. 2024) to account for varying results based on the exact context window slicing.

Table 3.2 is an example of using a context window of two seconds with 12 MFCCs results in training data of the following format:

$V1_1$	$V1_2$	...	$V1_{40}$	$V2_1$	...	$V12_{40}$	Time Start	Time End	Hump-back
1							0.0	1.0	
2							0.5	1.5	
3							1.0	2.0	
							...	...	
40							1.9	2.0	

Table 3.2: Table used for Training Data

Each row captures the evolutions of a MFCCs over the context window. $V1_1$ is the first MFCC of the time window starting at the value of “Time Start”. $V12_{40}$ is the value of the 12th MFCC, 40 time steps after the “Time Start” value. In general, V_{ij} is the value of the i th MFCC, j time steps after the “Time Start” value. Following this format $V_{ij} = V_{(i+1)_{j-1}}$ will be true for any $i = 1, \dots, 12$, and $j = 1, \dots, 40$.

Machine learning models are trained on these context windows.

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3.3.5 Model Assessing, Aggregation, and Thresholds

The initial predictions are based on the 0.1 second time windows. While these initial results are helpful and informative, predictions may be scattered and misleading. For example, a single positive prediction surrounded by extended periods of negative predictions is likely to truly be a false positive. An example of this is shown in figure 3.X. To avoid these issues, we use the same rolling evaluation windows as used in (Wong et al. 2025).

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3.4 Model Splitting

For training purposes, the annotated data was split into a training data set and an evaluation data set. The splitting process was not done at random. If splitting were done at random, portions of all whale calls are likely to be in the model development data. We want the model to be evaluated on data that has never been seen before. Thus, development data will be the first two hours of manual annotation, and the third hour will be the evaluation data set.

3.5 Model Fitting and Tuning

Models were tested using linear discriminant analysis, k-nearest-neighbors, and random forest.

Linear discriminant analysis is a parametric model. The model creates a linear boundary that maximizes the class separation. It operates under the assumptions that explanatory variable points are normally distributed, and that each class has the same covariance structure.

K-nearest-neighbors is a non-parametric method. The class of a prediction based on the majority class of its k closest neighbors in terms of distance. The distance metric can be customized, but it is typical to use the Euclidean distance. The normalization of coefficients ensures that the scale of an MFCC does not influence the distance between points.

Random forest is another non-parametric method. These forests are constructed of multiple decision trees. A decision tree is made up of multiple splits, usually multileveled. The metric that determines the split is the slicing that will increase the purity of the nodes the most.

In other words, the split is the decision that creates the most homogeneous child nodes (subgroups). This splitting occurs until the specified minimum tree size is reached, or the specified tree depth is reached. A random forest constructs many trees, but leaves out a random subset of variables in each tree. The prediction class of an unseen observation is determined by the majority class, voted on by the individual decision trees.

Model tuning involves tuning the appropriate model parameters to improve the results. K-fold cross-validation is a common machine model tuning method. First, training data is split into k equally sized (or as equal as possible) folds. For $i = 1, \dots, k$, fold i is held out, a model is created using all of the other folds, then creates predictions on fold i. The desired metrics of interest are calculated on these predictions.

Cross-validation provides a sense of how well the model will perform when given new data. The model created using k-1 folds will be similar to the model created using the entire training data set. Common values of k are 5 and 10. We use ten-fold cross-validation to tune these models.

3.6 Evaluation Metrics

The results of this binary classifier can be shown nicely in a confusion matrix; a 2 x 2 table showing the predicted results against the actual results. A confusion matrix is shown below in Figure 3.X. The rows of the table represent the predicted value, while the columns represent the true value. The four categories are true positive (TP), false positive (FP), true negative (TN), and false negative (FN). In context, true positives are correctly predicted whale calls, false positives are incorrectly predicted lack of whale calls, true negatives are correctly predicted lack of a call, and false negatives are incorrectly predicted lack of a call. The ideal model contains

only TP and FP. From a confusion matrix, general measures are calculated to understand a predictive model's strengths, weaknesses, and overall prediction rate. All metric values are proportion, and thus range from 0 to 1.

VISUAL OF EXAMPLE CONFUSION MATRIX

Accuracy is the overall correct prediction rate. The equation is:

$$\text{Accuracy} = (\text{TP} + \text{FP}) / (\text{TP} + \text{FP} + \text{TN} + \text{FN}) \quad (2)$$

where the denominator equals the total sample size, N.

Recall and Precision are two common metrics extracted from the confusion matrix.

Recall is the true positive rate. In context, the true positive rate is the proportion of actual whale calls that are predicted to be whale calls. Precision is the rate at which positive predictions are correct. Recall is incredibly important in the context of whale detection, as marine scientists want to minimize the amount of important whale call data overlooked. However, an effective machine learning model needs to be good at predicting periods of no calls. For example, a model that predicts everything to be a whale call will have a recall score of 1, despite the massive amount of overprediction. The precision score of a model is a measure of a model's overpredictions. In context, precision is the proportion of predicted whale calls that are actually whale calls. The equations for recall and precision are:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (3)$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (4)$$

F1-Score is a combination of recall and precision. The F1-score equation is

$$\text{F1} = (2 \times \text{Recall} \times \text{Precision}) / (\text{Recall} + \text{Precision}) \quad (5)$$

Visualizations of the interaction between precision and recall are helpful in finding a good balance between the two metrics. Precision-recall (PR) curves visualize a model's

predictive power at different thresholds. Figure 3.X shows an example PR curve. Precision on the y-axis is plotted against the recall on the x-axis. Models with low thresholds are more willing to make positive predictions, increasing recall. Models with high thresholds are less willing to make positive predictions, which tends to increase precision. Ideal models occupy the upper-right area of the graph.

VISUALIZATION OF EXAMPLE PR CURVE

Hildebrand et al. 2022 discussed how metrics can be calculated using two different manners in the cases of signal detection. The first method focuses on the predictions on the created time windows. After aggregation, each window is predicted to contain a whale call or not contain a whale call.

The second method focuses on the whale call unit. All positively predicted whale call windows are returned by the model. The amount of true whale call units that overlap with a predicted window is the number of true positives. False positives occur when a predicted whale call window does not overlap with a true whale call. False negatives occur when a true whale call does not have any predictions overlapping. This model cannot quantify a true negative, as “a finite time window is needed during which both the actual and the predicted detection were absent” (Hildebrand et al. 2022). Since neither precision nor recall care about the amount of true negatives, the PR curve is still usable. Thus, PR curves can be visualized for both the time window predictions, and individual whale call predictions.